

Topic: Scalable Enterprise Architectural Deployments of Agentic AI Solutions

Course Duration: 15 Hours

Course Type: Industry Oriented Course

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Pre-requisites

- Basic Python Programming
- Fundamentals of Machine Learning and Generative AI
- API Development Concepts
- Cloud Computing Fundamentals (AWS/Azure/GCP)
- Microservices and Containerization Basics
- Understanding of Enterprise Software Architecture

Course Objectives

Upon completion of this course, participants will be able to:

- Understand Agentic AI architecture patterns.
- Design scalable enterprise-grade AI agent ecosystems.
- Deploy multi-agent systems using modern frameworks.
- Implement secure, governed, and compliant AI solutions.
- Integrate Agentic AI with enterprise applications and databases.
- Monitor, optimize, and scale AI deployments in production.
- Build an end-to-end enterprise Agentic AI solution.

Lesson Path

S.No	Module	Hours
1	Foundations of Agentic AI & Enterprise Architecture	2
2	Enterprise AI Agent Design Patterns	2
3	LLMs, Tools, Memory & Knowledge Integration	2
4	Multi-Agent Systems & Orchestration Frameworks	2
5	Enterprise Integration Architecture	2
6	Scalability, Deployment & Cloud Architecture	2
7	Security, Governance & Responsible AI	1.5
8	Observability, Monitoring & Optimization	1
9	Capstone Project: Enterprise Agentic AI Deployment	0.5
Total		15 Hours

Detailed Module Coverage

Module 1: Foundations of Agentic AI & Enterprise Architecture (2 Hours)

Topics Covered

- Evolution from AI Assistants to AI Agents
- Agentic AI Concepts and Terminologies
- Enterprise AI Adoption Trends
- Autonomous vs Human-in-the-Loop Systems
- Enterprise Architecture Fundamentals
- AI Reference Architecture Layers
- Agent Lifecycle Management

Hands-on

- Exploring Agentic AI Use Cases across Industries

Module 2: Enterprise AI Agent Design Patterns (2 Hours)

Topics Covered

- Single-Agent Architecture
- Supervisor-Agent Pattern
- Planner-Executor Pattern
- Reflection and Self-Correction Pattern
- Human Approval Workflows
- Event-Driven Agent Architecture
- Workflow-based Agents

Hands-on

- Designing Enterprise Agent Workflows
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Module 3: LLMs, Tools, Memory & Knowledge Integration (2 Hours)

Topics Covered

- Enterprise LLM Selection Strategy
- Prompt Engineering for Agents
- Tool Calling and Function Invocation
- Retrieval Augmented Generation (RAG)
- Vector Databases
- Long-term and Short-term Memory
- Knowledge Graph Integration
- Enterprise Knowledge Repositories

Hands-on

- Building an Agent with Tool Calling and RAG
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Module 4: Multi-Agent Systems & Orchestration Frameworks (2 Hours)

Topics Covered

- Multi-Agent Collaboration Models
- Agent Communication Protocols
- Task Planning and Delegation
- Agent Orchestration Frameworks
- CrewAI Architecture
- LangGraph Fundamentals
- Microsoft AutoGen
- Agent-to-Agent (A2A) Communication

Hands-on

- Creating a Multi-Agent Business Workflow
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Module 5: Enterprise Integration Architecture (2 Hours)

Topics Covered

- Integrating Agents with Enterprise Systems
- APIs and Microservices
- ERP Integration
- CRM Integration
- SQL and NoSQL Databases
- Event Streaming Platforms
- Enterprise Service Bus (ESB)
- Enterprise Data Governance

Hands-on

- Agent Integration with Enterprise Database and APIs
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Module 6: Scalability, Deployment & Cloud Architecture (2 Hours)

Topics Covered

- Cloud-Native AI Architecture
- Kubernetes for Agentic AI
- Containerization using Docker
- Distributed Agent Deployment
- Serverless AI Patterns

- Load Balancing Strategies
- High Availability and Fault Tolerance
- Cost Optimization Strategies

Hands-on

- Deploying Agents on Kubernetes
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Module 7: Security, Governance & Responsible AI (1.5 Hours)

Topics Covered

- Enterprise AI Security Frameworks
- Authentication and Authorization
- Secrets Management
- AI Governance Frameworks
- Regulatory Compliance
- Data Privacy and Protection
- Responsible AI Principles
- Risk Management

Hands-on

- Implementing Secure Agent Access Controls
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Module 8: Observability, Monitoring & Optimization (1 Hour)

Topics Covered

- Agent Monitoring Metrics
- Tracing Agent Actions
- Observability Frameworks
- Performance Benchmarking
- Cost Monitoring
- Agent Evaluation Techniques
- Continuous Improvement Loops

Hands-on

- Building Agent Monitoring Dashboards
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Module 9: Capstone Project – Enterprise Agentic AI Deployment (0.5 Hour)

Project Scenario

Enterprise AI Copilot for Operations Management

Participants will build a scalable Agentic AI solution that:

- Accepts natural language business queries
- Retrieves information from enterprise databases
- Uses multiple collaborating agents
- Generates insights and recommendations
- Implements governance and security controls
- Deploys on cloud-native infrastructure

Deliverables

- Architecture Diagram
 - Agent Workflow Design
 - Deployment Strategy
 - Security Model
 - Monitoring Dashboard Design
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Recommended Technology Stack

Category	Tools
LLM Platforms	OpenAI GPT, Azure OpenAI, Claude, Gemini
Agent Frameworks	LangGraph, CrewAI, AutoGen
Vector DB	Pinecone, Chroma, Weaviate
Orchestration	Airflow, Temporal
Containers	Docker

Category	Tools
Container Platform	Kubernetes/OpenShift
Cloud	Azure, AWS, GCP
Monitoring	LangSmith, Prometheus, Grafana
Security	Azure Key Vault, HashiCorp Vault
CI/CD	GitHub Actions, Azure DevOps

Tentative Schedule

Date	Session	Topics Covered
Day 1	Session 1	Foundations of Agentic AI & Enterprise Architecture
	Session 2	Enterprise AI Agent Design Patterns
Day 2	Session 3	LLMs, Tools, Memory & Knowledge Integration
	Session 4	Multi-Agent Systems & Orchestration Frameworks
Day 3	Session 5	Enterprise Integration Architecture
	Session 6	Scalability, Deployment & Cloud Architecture
Day 4	Session 7	Security, Governance & Responsible AI
	Session 8	Observability, Monitoring & Optimization
	Session 9	Capstone Project & Presentations

Expected Learning Outcomes

Participants will be able to architect, deploy, govern, monitor, and scale enterprise-grade Agentic AI solutions using modern AI frameworks, cloud-native platforms, enterprise integration patterns, and responsible AI practices suitable for real-world production environments.